

STA 130A: Mathematical Statistics: Brief Course

Lecture 7: Joint, Marginal, and Conditional Distributions

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Fall 2026, UC Davis

Announcements

Homework 2 is due Thu, Oct 8, 11:59 PM PT

- Please submit on time and follow the submission instructions
 - To Gradescope via Canvas in a single PDF; select appropriate pages for each problem
 - Email submissions are not accepted

If you have questions or need help

- Instructor office hours: Wed, 2–3 PM (MSB 1143)
- TA office hours: Thu, 9–11 AM (MSB 1117)
- Feel free to ask questions during lecture, in office hours or post them on Piazza
 - Report in-class participation using the "[Participation in lectures](#)" survey on Canvas

* **After class:** Review the full slide decks, including examples and proofs not covered in class

Agenda

Last time:

- Functions of random variables
- Expectation and variance

Today:

- Spillover from Lecture 6: expected payoff and variability
- Joint and marginal PMFs
- Functions of multiple random variables and their expectations
- Conditional PMFs

Recap: Expectation

Basic definitions:

$$\mathbb{E}[X] = \sum_x x p_X(x),$$

$$\text{Var}(X) = \mathbb{E} \left[(X - \mathbb{E}[X])^2 \right].$$

Useful identities:

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x),$$

$$\mathbb{E}[aX + b] = a \mathbb{E}[X] + b,$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

- In general, $\mathbb{E}[g(X)]$ need not equal $g(\mathbb{E}[X])$; affine functions are an exception
- $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

Completing Lecture 6: Expected payoff and variability

Expected value is often used to compare decisions with random payoffs

Example (Choosing between two lotteries)

Suppose you must choose between options A and B . Let X_A and X_B denote the payoffs, in dollars:

$$X_A = 50 \text{ for sure,} \quad X_B = \begin{cases} 100, & \text{with probability 0.6,} \\ 0, & \text{with probability 0.4.} \end{cases}$$

Then their expected payoffs and variances (measured in dollars squared) are

$$\begin{aligned} \mathbb{E}[X_A] &= 50, & \mathbb{E}[X_B] &= 100 \times 0.6 = 60, \\ \text{Var}(X_A) &= 0, & \text{Var}(X_B) &= 100^2 \times 0.6 - 60^2 = 2400. \end{aligned}$$

- The mean measures average payoff: maximizing expected payoff favors B
- However, B has a 40% chance of paying nothing: variance measures variability, an aspect of risk

Why joint modeling?

Probabilistic models may involve several variables

- A medical diagnosis may involve several tests
- An investor may track the prices of multiple assets

These random variables are associated with the same experiment, sample space, and probability law, and their values may relate in interesting ways

Marginal (individual) PMFs alone do not describe how variables move together

Example

Example: Let $X, Y \in \{0, 1\}$, with both marginally Bernoulli(1/2).

- Model 1: $Y = X$. Then $P(X = Y) = 1$.
- Model 2: $Y = 1 - X$. Then $P(X = Y) = 0$.

Both models have the same individual PMFs for X and Y , but very different joint behavior

Joint probability mass function

Definition (Joint PMF)

If X and Y are discrete random variables, their **joint PMF** is

$$p_{X,Y}(x,y) = P(X = x, Y = y), \quad x, y \in \mathbb{R}.$$

- $p_{X,Y}(x,y) \geq 0$ for all x, y
- $\sum_x \sum_y p_{X,Y}(x,y) = 1$
- For any set $A \subseteq \mathbb{R}^2$, $P((X, Y) \in A) = \sum_{(x,y) \in A} p_{X,Y}(x,y)$

Example (Continued from the previous slide)

$p_{X,Y}(x,y)$	$y = 0$	$y = 1$
$x = 0$	1/2	0
$x = 1$	0	1/2

Model 1: $Y = X$

$p_{X,Y}(x,y)$	$y = 0$	$y = 1$
$x = 0$	0	1/2
$x = 1$	1/2	0

Model 2: $Y = 1 - X$

Marginal probability mass functions

The marginal PMFs are obtained by summing out the other variable:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y)$$

- Fix x : the events $\{X = x, Y = y\}$ partition $\{X = x\}$ as y varies
- With x labeling rows and y labeling columns, row sums give p_X ; column sums give p_Y
- Marginals describe each variable separately, not how the two variables vary together

Example (Check the two preceding tables)

In *both* models,

$$p_X(0) = p_{X,Y}(0, 0) + p_{X,Y}(0, 1) = \frac{1}{2},$$
$$p_Y(1) = p_{X,Y}(0, 1) + p_{X,Y}(1, 1) = \frac{1}{2}.$$

Thus both have Bernoulli(1/2) marginals, despite their different joint PMFs.

Example: Joint PMF and marginal PMF

Example

$p_{X,Y}(x,y)$	$y = 1$	$y = 2$	$y = 3$	$y = 4$	$p_X(x)$
$x = 1$	1/20	1/10	3/20	0	3/10
$x = 2$	0	1/20	1/10	1/5	7/20
$x = 3$	1/10	1/10	1/10	1/20	7/20
$p_Y(y)$	3/20	1/4	7/20	1/4	1

The marginal PMFs are given by the row sums and the column sums

We can also compute probabilities of events involving both variables, for example:

$$\begin{aligned}P(X = Y) &= p_{X,Y}(1,1) + p_{X,Y}(2,2) + p_{X,Y}(3,3) \\ &= 1/5,\end{aligned}$$

$$\begin{aligned}P(X < Y) &= p_{X,Y}(1,2) + p_{X,Y}(1,3) + p_{X,Y}(1,4) + p_{X,Y}(2,3) + p_{X,Y}(2,4) + p_{X,Y}(3,4) \\ &= 3/5.\end{aligned}$$

Function of multiple random variables

A function $Z = g(X, Y)$ defines another random variable

- Its PMF can be calculated from the joint PMF by

$$p_Z(z) = \sum_{\{(x,y) | g(x,y)=z\}} p_{X,Y}(x,y)$$

- Different pairs may produce the same value of $g(X, Y)$; combine all such cells

To compute the expectation, we can use the joint PMF directly or use p_Z :

$$\begin{aligned} \mathbb{E}[g(X, Y)] &= \sum_{x,y} g(x,y) p_{X,Y}(x,y) \\ &= \sum_z \underbrace{g(x,y)}_z \underbrace{\sum_{(x,y): g(x,y)=z} p_{X,Y}(x,y)}_{p_Z(z)} = \sum_z z p_Z(z) \end{aligned}$$

Linearity of expectation (no independence required): If g is an affine function,

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$$

Example: Distribution and mean of $Z = X + Y$

Example

Let $Z = X + Y$ where

$p_{X,Y}(x,y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	$1/8$	$1/8$	$1/4$
$x = 1$	$1/4$	$1/8$	$1/8$

Then

$$P(Z = 0) = 1/8,$$

$$P(Z = 1) = 1/8 + 1/4 = 3/8,$$

$$P(Z = 2) = 1/4 + 1/8 = 3/8,$$

$$P(Z = 3) = 1/8.$$

Hence

$$\mathbb{E}[Z] = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{3}{2}.$$

Alternatively, linearity gives the same mean from the marginal PMFs:

$$\mathbb{E}[X] + \mathbb{E}[Y] = \frac{1}{2} + 1 = \frac{3}{2} = \mathbb{E}[Z].$$

Pop-up quiz

Using the joint PMF table from the previous example,

$p_{X,Y}(x,y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	$1/8$	$1/8$	$1/4$
$x = 1$	$1/4$	$1/8$	$1/8$

which statement is correct?

- A) $p_X(0) = 3/8$
- B) $p_Y(2) = 3/8$ and $P(X < Y) = 1/2$
- C) $p_{X,Y}(0,2) = p_X(0)p_Y(2)$
- D) $\mathbb{E}[X + Y] = 1$

Answer: B.

The column sum gives $p_Y(2) = 3/8$, and summing the three cells with $x < y$ gives $1/2$.

Self-study: More than two random variables

For three discrete random variables, the joint PMF is defined analogously as

$$p_{X,Y,Z}(x,y,z) = P(X=x, Y=y, Z=z), \quad \forall (x,y,z) \in \mathbb{R}^3$$

Marginal PMFs are obtained by summing over coordinates we do not retain:

$$p_{X,Y}(x,y) = \sum_z p_{X,Y,Z}(x,y,z), \quad p_X(x) = \sum_y \sum_z p_{X,Y,Z}(x,y,z).$$

Example (The preceding sum example, $Z = X + Y$)

Only triples with $z = x + y$ can occur. For example,

$$p_{X,Y,Z}(0,2,2) = \frac{1}{4}, \quad p_{X,Y,Z}(0,2,1) = 0 \quad \implies \quad p_{X,Y}(0,2) = \sum_{z=0}^3 p_{X,Y,Z}(0,2,z) = \frac{1}{4}.$$

More generally, for n discrete random variables $X_1, \dots, X_n$,

$$p_{X_1, \dots, X_n}(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n), \quad \forall (x_1, \dots, x_n) \in \mathbb{R}^n$$

Self-study: Expectations of several random variables

The same weighted-average rule applies to n discrete random variables:

$$\mathbb{E}[g(X_1, \dots, X_n)] = \sum_{x_1, \dots, x_n} g(x_1, \dots, x_n) p_{X_1, \dots, X_n}(x_1, \dots, x_n).$$

Linearity of expectation

For constants $a_1, \dots, a_n$,

$$\mathbb{E} \left[\sum_{i=1}^n a_i X_i \right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

Note: independence is not required

- The finite-sum rule follows by repeated application of two-variable linearity
- To find the mean of a sum, we need only the individual means, not the full joint PMF

Example: Mean of the binomial

Example (Mean of a binomial random variable)

Let X count successes in n independent trials, each with success probability p , so $X \sim \text{Binomial}(n, p)$. Write

$$X = X_1 + \cdots + X_n,$$

where

$$X_i = \begin{cases} 1, & \text{if trial } i \text{ is a success,} \\ 0, & \text{otherwise.} \end{cases}$$

Then each $X_i \sim \text{Bernoulli}(p)$, so $\mathbb{E}[X_i] = p$.

By linearity of expectation,

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n] = np.$$

Example: The hat problem

Example (Expected number of people who get their own hats back)

Suppose $n \geq 1$ people put their hats into a box, and then each person picks one hat at random.

Question: What is $\mathbb{E}[X]$, where X is the number of people who get back their own hat?

Computing p_X looks complicated... but for each $i = 1, \dots, n$, we may define

$$X_i = \begin{cases} 1, & \text{if person } i \text{ gets back their own hat,} \\ 0, & \text{otherwise.} \end{cases}$$

Observe that $P(X_i = 1) = 1/n$, so $\mathbb{E}[X_i] = 1/n$.

Since $X = X_1 + \dots + X_n$, by linearity of expectation,

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X_i] = n \cdot \frac{1}{n} = 1.$$

Although the number of correct matches is random, its expected value is always 1, regardless of n .

Writing complicated counts as sums of indicators may make $\mathbb{E}[X]$ easy to compute

Conditioning a random variable on an event

Definition

The **conditional PMF** of a random variable X given an event A with $P(A) > 0$ is

$$p_{X|A}(x) = P(X = x \mid A) = \frac{P(\{X = x\} \cap A)}{P(A)}$$

- This is the same conditioning idea as before, now applied to the event $\{X = x\}$
- For each fixed event A with $P(A) > 0$, $p_{X|A}$ is a valid PMF

Example

Let X be the roll of a fair die, and let A be the event that the roll is an even number

$$p_{X|A}(x) = \frac{P(X = x \text{ and } X \text{ is even})}{P(\text{roll is even})} = \begin{cases} 1/3, & \text{if } x \in \{2, 4, 6\}, \\ 0, & \text{otherwise.} \end{cases}$$

Conditioning keeps only the probability mass compatible with A , then renormalizes it.

Conditioning a random variable on another

Let X, Y be random variables associated with the same experiment

Knowing $Y = y$ may provide partial knowledge about the value of X

Definition

For any y with $p_Y(y) > 0$, the **conditional PMF** of X given $Y = y$ is

$$p_{X|Y}(x | y) = P(X = x | Y = y)$$

- This is a specialization of the previous definition applied to the event $A = \{Y = y\}$

$$p_{X|Y}(x | y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

- For each fixed y with $p_Y(y) > 0$, the function $x \mapsto p_{X|Y}(x | y)$ is a valid PMF
- Operationally: take the slice $Y = y$ of the joint table and renormalize it

Illustration: Conditional PMF from a joint table

Example

Suppose the joint PMF of X and Y is

$p_{X,Y}(x,y)$	$y = 0$	$y = 1$	$y = 2$	$p_X(x)$
$x = 0$	$1/8$	$1/8$	$1/4$	$1/2$
$x = 1$	$1/4$	$1/8$	$1/8$	$1/2$
$p_Y(y)$	$3/8$	$1/4$	$3/8$	

Condition on the event $\{Y = 2\}$. Since $p_Y(2) = 3/8$,

$$p_{X|Y}(0 | 2) = \frac{1/4}{3/8} = \frac{2}{3}, \quad p_{X|Y}(1 | 2) = \frac{1/8}{3/8} = \frac{1}{3}.$$

Given $Y = 2$, the conditional PMF of X is

$$p_{X|Y}(x | 2) = \begin{cases} 2/3, & x = 0, \\ 1/3, & x = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Illustration: Building a joint PMF sequentially

Example (Short Q&A session)

Let X be the number of questions asked in a short Q&A session, and assume

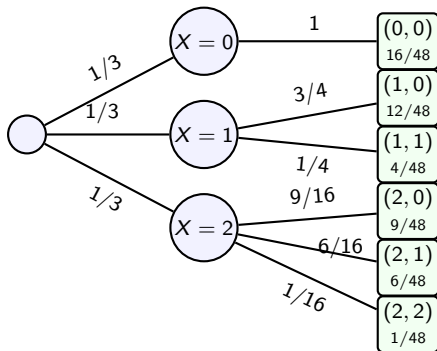
$$P(X = 0) = P(X = 1) = P(X = 2) = \frac{1}{3},$$

Suppose that given $X = x$, each question is answered incorrectly with probability $1/4$, independently of the others. Let Y be the number answered incorrectly. Thus

$$Y \mid X = x \sim \text{Binomial}(x, 1/4),$$

and therefore,

$$p_{X,Y}(x, y) = p_X(x) p_{Y|X}(y \mid x).$$



$p_{X,Y}(x, y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	16/48	0	0
$x = 1$	12/48	4/48	0
$x = 2$	9/48	6/48	1/48

Pop-up quiz

Using the joint PMF table

$p_{X,Y}(x,y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	$1/8$	$1/8$	$1/4$
$x = 1$	$1/4$	$1/8$	$1/8$

which statement is correct?

- A) $P(X = 1 \mid Y = 0) = 1/3$
- B) $P(X = 1 \mid Y = 2) = 1/3$
- C) $P(X = 1 \mid Y = 2) = 1/2$
- D) $P(X = 1 \mid Y = 0) = P(X = 1)$

Answer: B.

Because $P(Y = 2) = 3/8$, so

$$P(X = 1 \mid Y = 2) = \frac{P(X = 1, Y = 2)}{P(Y = 2)} = \frac{1/8}{3/8} = \frac{1}{3}.$$

Wrap-up

Joint and marginal distributions

- The joint PMF describes (X, Y) ; sum cells for events and rows/columns for marginals
- Marginal PMFs alone do not generally determine the joint law

Functions of several random variables

- Group equal values of $g(x, y)$ to obtain its PMF, or calculate its mean directly:

$$\mathbb{E}[g(X, Y)] = \sum_{x,y} g(x, y) p_{X,Y}(x, y)$$

- Expectations of finite sums add, even without independence

Conditional PMFs: restrict and renormalize

$$p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}, \quad p_Y(y) > 0.$$

Suggested reading: [BT08, Ch. 2.5–2.6]

Next lecture: Conditional means, total expectation, independence of random variables

References



Dimitri Bertsekas and John N Tsitsiklis.

Introduction to probability, volume 1.

Athena Scientific, 2nd edition, 2008.